Particular Solutions

Exam code: 9MA0

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Particular solutions

What is a particular solution?

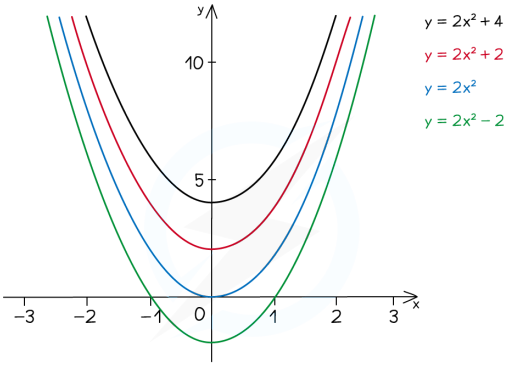
- Ensure you are familiar with **General Solutions** first
- With extra information, the constant of integration, **c**, can be found
- This means the **particular solution** (from the family of solutions) can be found

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



USING (0, 4) FIND c

AT (0, 4): $4 = 2 \times 0^2 + c$

$c = 4$

∴ $y = 2x^2 + 4$ ← PARTICULAR SOLUTION

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What is a boundary condition and an initial condition?

- A **boundary condition** is a piece of extra information that lets you find the particular solution
  - For example knowing **y = 4** when **x = 0** in the preceding example
  - In a model this could be a particle coming to rest after a certain time, ie **v = 0** at time **t**

e.g. FIND THE PARTICULAR SOLUTION OF THE DIFFERENTIAL EQUATION

$$4 - \frac{dv}{dt} = 2$$

GIVEN THAT $v = 0$ WHEN $t = 8$

$\frac{dv}{dt} = 2$

$v = 2t + c$ ← GENERAL SOLUTION

BOUNDARY CONDITION

AT (8, 0): $0 = 2 \times 8 + c$

$c = -16$

$t = 0$ (NOT $v = 0$) WOULD BE AN INITIAL CONDITION

∴ PARTICULAR SOLUTION IS $v = 2t - 16$

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- Differential equations are used in modelling, experiments and real-life situations
- A boundary condition is often called an **initial condition** when it gives the situation at the start of the model or experiment
  - This is often linked to time, so **t = 0**

GIVEN THAT AT $x = 0$, $y = 3$

$\frac{dy}{dx} = 6 - 2x$ ← REARRANGE

$y = \int (6 - 2x)\,dx$ ← (THIS LINE IS NOT ESSENTIAL)

$y = 6x - x^2 + c$ ← GENERAL SOLUTION

AT (0,3): $3 = 6 \times 0 - 0^2 + c$

$c = 3$

INITIAL CONDITION

∴ PARTICULAR SOLUTION IS

$y = 6x - x^2 + 3$

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- It is possible to have two boundary conditions
  - eg a particle initially at rest has velocity, **v = 0** and acceleration, **a = 0** at time, **t = 0**
  - for a **second order** differential equation you need **two** boundary conditions to find the particular solution

Worked Example

The velocity of a particle, initially at rest, is modelled by the differential equation

$$\frac{dv}{dt} = 2t - 2$$

where v is the velocity of the particle and t is time since the particle began moving.

(a) Find the velocity of the particle after 3 seconds

(b) According to this model when is the particle at rest (other than initially)?

a) $v = \int 2t - 2\,dt$ ← INTEGRATE TO FIND v

$v = t^2 - 2t + c$ ← GENERAL SOLUTION

GIVEN IN WORDS "INITIALLY AT REST"

AT $t = 0$, $v = 0$, ∴ $c = 0$

∴ PARTICULAR SOLUTION IS $v = t^2 - 2t$

∴ AT $t = 3$, $v = 3^2 - 2 \times 3$

$v = 3\ ms^{-1}$

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